
Amirkabir University of Technology
Course: Autonomous Aerial Mobile Robots
LiDAR ICP Project

Project Overview

This project focuses on LiDAR-based motion estimation for aerial robots. You will use Iterative Closest Point (ICP) to align individual 3D LiDAR scans and then extend the same idea to build a local point-cloud map from a sequence of scans.

The goal is not only to run an existing ICP function, but also to understand how scan spacing, preprocessing, initialization, and accumulated drift affect the quality of registration and mapping.

Dataset

Use a scan sequence derived from the [Oxford Osprey dataset](#), an aerial LiDAR–visual–inertial dataset collected with a UAV platform. The public Osprey release is distributed in ROS bag or ROS2 database form. If students cannot access or extract the data themselves, an instructor-prepared non-ROS subset will be provided as a zip file.

Tasks

- 1) **Single scan matching.** Select two LiDAR scans from the same flight with a small but visible baseline, for example scans separated by 1–3 seconds. Register the source scan to the target scan using ICP. Report the estimated rigid transformation, convergence behavior, fitness/RMSE or an equivalent registration metric, and visualizations before and after alignment. If your implementation supports both methods, compare point-to-point ICP with point-to-plane ICP.
- 2) **Consecutive matching and local mapping.** Register a sequence of LiDAR scans consecutively. Compose the pairwise transformations to estimate the UAV trajectory, transform each scan into a shared coordinate frame, and accumulate the aligned scans into a local 3D map. Visualize the reconstructed point-cloud map together with the estimated trajectory.
- 3) **Analysis.** Discuss the main sources of error in your result. Your discussion should address drift, scan spacing, voxel size, ICP initialization, local minima, and scene geometry.

Expected Submission

- Preprocessing settings, ICP parameters, stopping criteria, and initialization method.

- Single-pair registration results, including before/after alignment figures and numerical metrics.
- Consecutive registration results, including the accumulated map and estimated UAV trajectory.
- A short technical discussion explaining where the method succeeds, where it fails, and why.